

Project Overview

Nexus is a hybrid blockchain-based digital banking platform that combines decentralized finance with a traditional banking user experience. The goal of the project is to demonstrate how blockchain technology can be made accessible to everyday users by presenting it through a familiar and intuitive interface similar to modern fintech applications. The platform is built for the Indian financial context, displaying values in INR and incorporating concepts like UPI address support and compliance-aware design.

How the System Works

When a user opens the application they are presented with a login screen where they can either sign in using demo credentials or connect directly using their MetaMask wallet. Once authenticated, the user lands on a dashboard that displays their real Ethereum balance fetched from the Sepolia testnet, a dynamic spending trend chart built from their actual transaction history, and a list of recent transactions.

To send funds, the user goes through a three step flow where they enter a recipient address or select one from their saved address book, enter an amount in ETH which is converted to INR in real time, add an optional memo, and then confirm the transaction. At the confirmation step MetaMask opens and the user signs the transaction with their private key. The transaction is then broadcast to the Sepolia testnet and the result is saved locally so it appears in the dashboard history.

The receive page generates a real QR code from the connected wallet address which can be shared or copied. The security center allows users to manage session settings, toggle multi-signature authentication, and configure spending limits and low balance alerts. The address book lets users save frequently used wallet addresses with custom names and mark favourites for quick access during the send flow. An admin dashboard built with Streamlit provides a separate internal view with platform metrics, transaction volume charts, audit logs, and system status.

Stack Used

The frontend is built with React and Vite, styled using Tailwind CSS, and uses Ethers.js to interact with the user's MetaMask wallet and the Ethereum blockchain. The backend is a FastAPI application written in Python that handles API requests and connects to the blockchain using Web3.py. The database layer uses Supabase which provides a hosted PostgreSQL instance along with a REST API. The smart contract is written in Solidity and deployed on the Sepolia Ethereum testnet. The admin dashboard is built with Streamlit. Wallet authentication and transaction signing is handled entirely through MetaMask, meaning no private keys are ever stored or transmitted through the application.